B.Tech (ME)– IV-I Sem

23A03703d	**SMART MANUFACTURING (Professional Elective-V)**	**L**	**T**	**P**	**C**
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Course objectives: The objectives of the course are to	
1	**Understand** the fundamental concepts of Smart Manufacturing, Industry 4.0 and IoT in Manufacturing
2	**Understand** the fundamental concepts of Data Analytics and Artificial Intelligence in Manufacturing
3	Know about Smart Sensors, RFID, and Block chain in Manufacturing
4	**Explain the techniques of** Digital Twin & Simulation in Manufacturing
5	**Familiarize** Sustainable and Green Manufacturing

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Define and explain the evolution of Smart Manufacturing, Industry 4.0 and IoT in Manufacturing	L1,L2,L6
2	Concepts of Data Analytics and Artificial Intelligence in Manufacturing	L2,L4
3	Apply knowledge of Smart Sensors, RFID, and Block chain in Manufacturing	L3,L5,L6
4	**Summarise the techniques of** Digital Twin & Simulation in Manufacturing	L2,L3,L5,
5	Importance of Sustainable and Green Manufacturing.	L1,L3,L6

Unit – I

Introduction to Smart Manufacturing

Definition, Evolution, and Importance of Smart Manufacturing, Industry 4.0 and Its Impact on Manufacturing, Traditional vs. Smart Manufacturing, Key Technologies in Smart Manufacturing

Cyber-Physical Systems & Industrial IoT (IIoT)

Basics of Cyber-Physical Systems (CPS), Internet of Things (IoT) in Manufacturing, Smart Sensors and Actuators Communication Protocols (MQTT, OPC-UA, Modbus),

Unit – II Data Analytics and Artificial Intelligence in Manufacturing

Role of Big Data in Manufacturing, Machine Learning & Deep Learning for Smart Manufacturing, Predictive Maintenance and Anomaly Detection, Digital Twin Technology

Automation & Robotics in Smart Manufacturing

Industrial Automation: PLCs, SCADA, and DCS, Autonomous Robots and Cobots (Collaborative Robots), Additive Manufacturing (3D Printing), Case Studies of Smart Factories.

Unit – III
Cloud Computing and Edge Computing in Manufacturing

Introduction to Cloud Computing for Manufacturing, Edge and Fog Computing in Real-time Manufacturing Systems, Cloud-based Manufacturing Execution Systems (MES)

Smart Sensors, RFID, and Block chain in Manufacturing

Role of Smart Sensors and RFID in Inventory Management, Blockchain for Secure Supply Chains Smart Contracts and Decentralized Manufacturing Systems

Unit- IV

Digital Twin & Simulation in Manufacturing

Digital Twin Technology and Virtual Prototyping, Simulation Software for Smart Manufacturing, Augmented Reality (AR) and Virtual Reality (VR) in Industry 4.0

Cybersecurity in Smart Manufacturing

Threats and Risks in Smart Manufacturing, Cybersecurity Frameworks for Industrial Systems, Best Practices for Securing IoT and IIoT Devices

Unit – V

Sustainable and Green Manufacturing

Energy Efficiency in Smart Manufacturing, Sustainable Supply Chain Management, Role of AI in Green Manufacturing

Case Studies and Future Trends

Case Studies of Leading Smart Manufacturing Companies, Emerging Technologies in Smart Manufacturing, Future of Industry 5.0 and Beyond

Text Books:

1. Green Manufacturing Processes and Systems - J. Paulo Davim/Springer 2013
2. Smart manufacturing by Kamalakar Mutalik The Righte order (2023)

Reference Books:

1. Introduction To Smart Manufacturing And Automation Dr. Rajkumar. E Namya Press (29 March 2024); 213, Vardan Hosue 7/28 Ansari Road Daryganj Delhi 110002.
2. Smart Manufacturing by Michael Deng (Author), Colin Koh Kindle Edition (2023)

Online Learning Resources:

- https://www.youtube.com/watch?v=sdgI072DJNM&pp=ygUUU01BUlQgTUFOVUZBQ1RVUklORyA%3D(2023)